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Centrifugal confinement fusion thruster

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ABSTRACT

Centrifugal confinement fusion, a promising alternative to toroidal confinement devices like tokamaks and stellarators, leverages supersonic plasma rotation within a magnetic mirror configuration to achieve simplified coil design, compactness, and enhanced stability. This brief article explores the potential of centrifugal confinement fusion for propulsion applications. A previous concern regarding the escape of energetic ions, essential for propulsion, is addressed through test-particle simulations. The results indicate that the earlier estimate based on adiabatic invariance was overly pessimistic, and the underlying physics is clarified.

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1. Introduction

Centrifugal confinement, an approach for magnetic confinement fusion, utilizes supersonic plasma rotation to overcome the loss cone problem and flute interchange mode, two major obstacles of a simple mirror configuration (1, 2). The centrifugal force from the supersonic plasma rotation provides an effective gravitational potential well that closes the ion loss cone except for the energetic ions in the high-energy tail, and the associated velocity shear of the rotation stabilizes the flute interchange modes (3–7). The simplicity of coil design, compactness, and stability make centrifugal confinement an attractive alternative to toroidal confinement devices such as tokamaks and stellarators.

Recently, a concept of utilizing a centrifugal confinement fusion device for propulsion has been proposed. The idea is to make one of the mirrors weaker such that the energetic ions preferentially escape from the weaker mirror, thereby yielding a net propulsion (Figure 1(a)). For a reactor with one cubic meter of volume and a plasma density of $10^{20}/\text{m}^3$, the propulsion force is estimated to be 500N, which can send a spacecraft of 1000 kg from Earth to Jupiter in about one month (8).

However, a concern was raised regarding whether energetic ions can actually escape from the mirror machine (9). The argument goes as follows. Ions and electrons are held together by Coulomb interaction. Therefore, ions can escape only if electrons can escape. Due to their lighter mass, electrons are more tightly tied to magnetic field lines than ions. If the electrons remain tied to the field lines, electrons escaped from one end will circle around and return to the other end. If the electrons cannot escape, the energetic ions

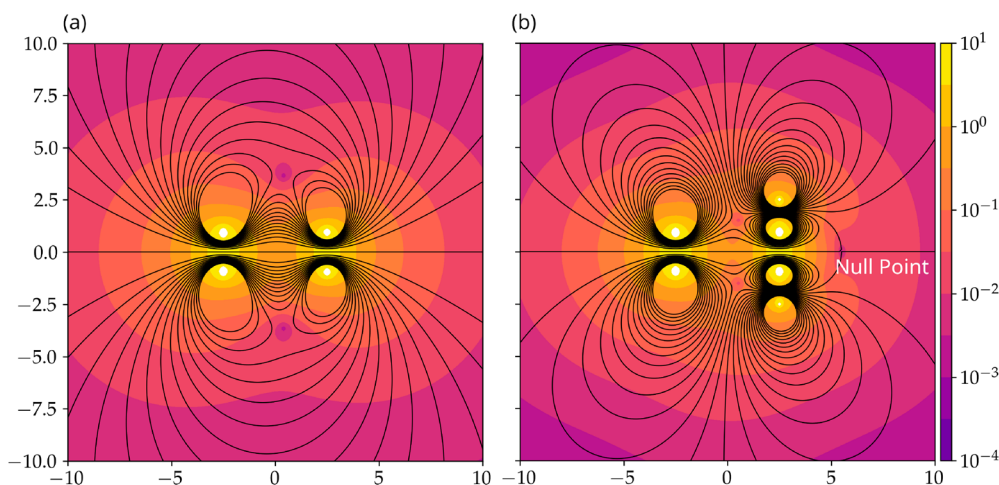


Figure 1. (a) An asymmetric magnetic mirror machine with the mirror coil on the right weaker than the one on the left. Energetic ions preferentially exhaust from the weaker mirror, yielding a net propulsion to the left. (b) An alternative asymmetric magnetic mirror machine with a third coil with a dipole moment that exactly cancels the dipole moments of the other two coils, making the global magnetic field is a quadrupole. The magnetic field also has a null point on the right hand side of the machine. Black lines are magnetic field lines, and color denotes the magnitude of magnetic field in an arbitrary unit.

will be constrained, preventing net propulsion. One may estimate at what distance the electrons can break free from the field line via the adiabatic invariance of the magnetic moment $\mu = mv_{\perp}^2/2B$. Let r be the distance from the mirror machine. A configuration like Figure 1(a) is a dipole field at large scales, for which the magnetic field decreases with the distance r as $B \propto 1/r^3$. Therefore, the perpendicular speed $v_{\perp} \propto r^{-3/2}$ and the electron Larmor radius $\rho_e \propto v_{\perp}/B \propto r^{3/2}$. The adiabatic invariance of μ requires $\rho_e |\nabla B| \ll B$. Conservatively, the distance where the electrons become demagnetized may be estimated by the condition $\rho_e |\nabla B| \sim B$. Let ρ_{e0} be the electron Larmor radius inside the mirror machine and a be the characteristic size of the machine. From $\rho_e \sim \rho_{e0}(r/a)^{3/2}$ and $B/|\nabla B| \sim r$, the condition $\rho_e |\nabla B| \sim B$ yields $r/a \sim (a/\rho_{e0})^2$. For a reactor with the temperature $\sim 10\text{keV}$ and the size $a \sim 1\text{m}$, the ratio $a/\rho_{e0} \sim 10^4$; therefore, the electrons are demagnetized at $r/a \sim 10^8$, which is enormous. Although this estimate is conservative (e.g. the requirement of $\rho_e \sim B/|\nabla B|$ may be excessive and collisions are neglected) and the electrons may demagnetize at a shorter distance, the general idea suggests that ions may be constrained by the electrons, limiting net propulsion.

To overcome this problem, an alternative magnetic coil configuration, shown in Figure 1(b), has been proposed (10). In this new configuration, the original two mirror coils are of the same strength, but a third coil of a larger radius is added next to the coil on the right. The electric current in the third coil is opposite to the current in the first two coils, making the mirror on the right weaker. Moreover, the current in the third coil is adjusted to cancel the net dipole moment of the first two coils, making the global magnetic field a quadrupole. Because the quadrupole field decreases with r as $B \propto 1/r^4$, repeating the estimate for a quadrupole field yields a condition for electron detachment at $r/a \sim a/\rho_{e0} \sim 10^4$, which is significantly less stringent than the previous estimate of 10^8 for a dipole field. An additional

advantage of this new configuration is the presence of a null point on the right. Because electrons that get sufficiently close to the null point will be demagnetized, the null point should also help the detachment of electrons.

2. Test particle simulations

Given the estimates in the previous section, the key questions are: Is it really very difficult for electrons to escape from a dipole field as the estimate suggests? If that is the case, can the alternative configuration of a quadrupole field come as a rescue? To address these questions, test-particle simulations were performed in both dipole and quadrupole fields with a Boris solver. For simplicity, only the magnetic field is considered, and the electric field associated with the plasma rotation is ignored.

The simulations show that electrons readily escape the mirror machine after passing the mirror throat, even in a dipole field. Figure 2 illustrates the trajectories, the magnetic moments μ with the distance r , and the electron Larmor radius ρ_e with the distance r for a sample of electrons in both dipole and quadrupole configurations. These results indicate that the magnetic moment conservation breaks down beyond $r \sim 100$, and the electron Larmor radius ρ_e becomes comparable to the distance r from the mirror at $r \sim 1000$. Interestingly, there is no significant difference between the dipole and the quadrupole field results, with the quadrupole field slightly accelerating the breakdown of magnetic moment conservation.

What causes the conservation of magnetic moment to break down earlier than expected in a dipole field? Upon further reflection, it is realized that the adiabatic invariance of

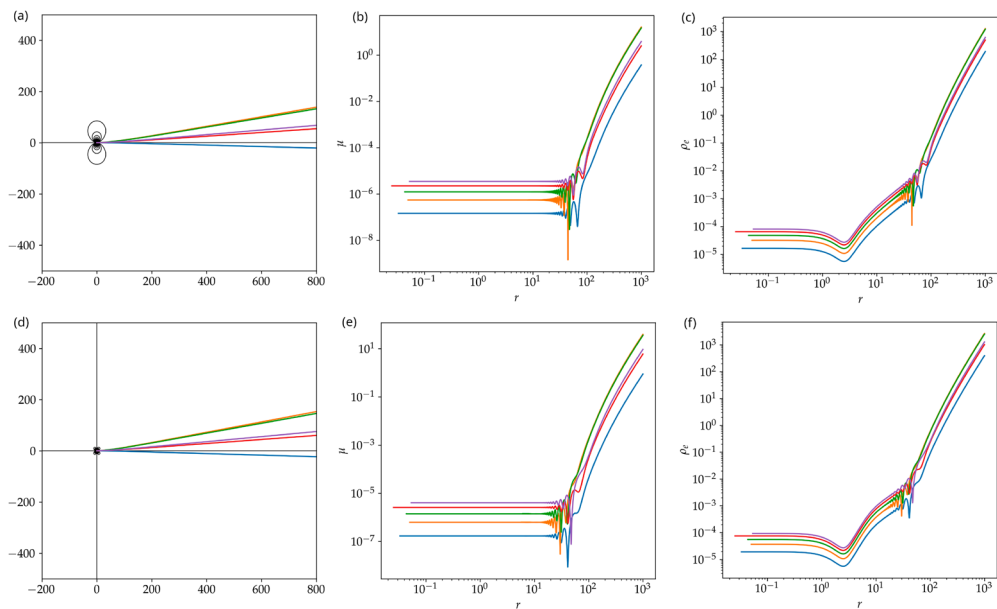


Figure 2. (a) The trajectories, (b) the magnetic moments μ versus the distance r , and (c) the Larmor radius ρ_e versus r for a sample of electrons in a dipole magnetic mirror machine. (d)–(f) The results for a quadrupole magnetic mirror machine with electrons of the same initial conditions.

μ requires not only $\rho_e |\nabla B| \ll B$ but also $dB/dt \ll \omega_{ce} B$, where ω_{ce} is the electron gyro-frequency. These two conditions are typically comparable if the parallel speed $v_{\parallel}$ and the perpendicular speed $v_{\perp}$ are comparable. In the present case, however, as electrons move away from the magnetic mirror, the perpendicular speed decreases as $v_{\perp} \propto r^{-3/2}$, whereas $v_{\parallel}$ increases to conserve the total energy. Therefore, the condition $dB/dt \ll \omega_{ce} B$ must be considered separately.

Qualitatively, we may estimate $dB/dt \sim v_{\parallel} |\nabla B|$, where $v_{\parallel}$ is on the order of the initial electron thermal speed v_0 . For a dipole field, the gyro-frequency $\omega_{ce} \sim v_{\perp} / \rho_e \sim (a/r)^3 v_0 / \rho_{e0}$. Hence, the condition $dB/dt \ll \omega_{ce} B$ yields $r/a \ll (a/\rho_{e0})^{1/2}$. This is a much weaker condition than the previous estimate of $r/a \ll (a/\rho_{e0})^2$ for a dipole field. The physical picture goes as follows. As electrons move away from the mirror, their gyro-period lengthens, allowing them to traverse significant distances within a single gyro-period. This leads to a significant change in the magnetic field within a gyro-period, causing the breakdown of the magnetic moment conservation. For a fusion reactor with $a/\rho_{e0} \sim 10^4$, this breakdown occurs at $r/a \sim 100$, aligning with the simulation results.

3. Conclusion

In conclusion, this study shows that electrons are not as strongly attached to the magnetic mirror dipole field as suggested by an earlier pessimistic estimate. The gap in the previous estimate regarding the escape condition has now been addressed and clarified. Consequently, the concern about constraint on energetic ion escape due to electrons, which could have limited net propulsion, is no longer a significant issue. The results also indicate that a quadrupole field may not be necessary, because a dipole field is already sufficient.

Centrifugal confinement, therefore, shows promise for both fusion power generation and space propulsion. Built upon the success of the Maryland Centrifugal Experiment (MCX) at the University of Maryland, College Park, the Centrifugal Mirror Fusion Experiment (CMFX) (11) at the University of Maryland, Baltimore County, has recently achieved significant advancements. These include steady-state operation for several seconds and the detection of neutrons from Deuterium-Deuterium fusion (12). The future of centrifugal confinement appears bright, with exciting developments expected from future generations of this technology.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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